

Index expectation

(V, E) finite simple graph
 $(\Omega, \mathcal{A}, P)$ probability space of coloring functions on V

$$K(v) = E[i_g(v)]$$

defines a curvature that satisfies Gauss-Bonnet:

$$\sum_{v \in V} K(v) = \chi(G)$$

Proof: take expectation of the Poincare-Hopf formula:

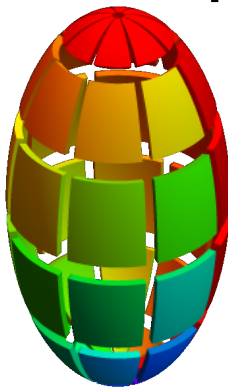
$$\sum_{v \in V} i_g(v) = \chi(G)$$

$$E[\sum_{v \in V} i_g(v)] = \chi(G)$$

and apply Fubini

$$\sum_{v \in V} E[i_g(v)] = \chi(G)$$

Example:



$$r(s, t, u, v) = \begin{bmatrix} 2 \cos(t) \\ \cos(s) \sin(t) \\ \sin(s) \sin(t) \cos(u) \\ \sin(s) \sin(t) \sin(u) \cos(v) \\ \sin(s) \sin(t) \sin(u) \sin(v) \end{bmatrix}$$

4 dim ellipsoid
 $M \subset \mathbb{R}^5$

$$K(x) = -\frac{128}{(5 - 3 \cos(2t))^2 (3 \cos(2t) - 5)}$$

$$dV = \frac{\sqrt{\sin^4(s) \sin^6(t) (- (3 \cos(2t) - 5)) \sin^2(u)}}{\sqrt{2}}$$

Gauss-Bonnet-Chern

$$\int_M K(x) dV(x) = 2$$

Continuum

(M, g) compact Riemannian manifold
 $(\Omega, \mathcal{A}, P)$ probability space of Morse functions on M

$$K(v) = E[i_g(v)]$$

satisfies Gauss-Bonnet. For nice probability spaces, it is a smooth function on M

$$\int_M K(v) dV = \chi(M)$$

Rewrite Poincare -Hopf:

as $\sum_{v \in M} i_g(v) = \chi(M)$

$$\int_M i_g(v) = \chi(M)$$

where now,

i_g divisor (a signed point measure).

Now apply expectation and Fubini.



Gauss Bonnet Chern

(M, g) Riemannian, compact even dimensional

$$\int_M \sum_{\pi} \frac{\prod_{k=1}^d K_{\pi(2k-1), \pi(2k)}}{d! (4\pi)^d} dV = \chi(M)$$

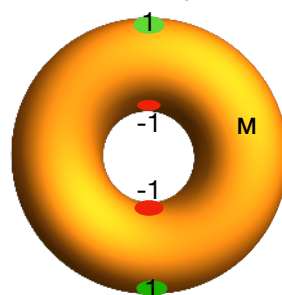
Weyl and Gilkey:

The GBC integrand is the unique local frame invariant curvature.

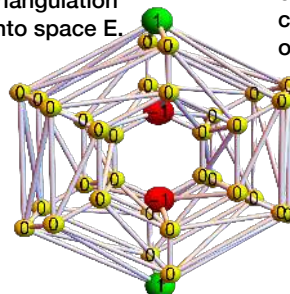
The index expectation coming from a Nash embedding into an Euclidean space must be the Gauss-Bonnet-Chern curvature!

Discrete-Continuum

Nash embed manifold M into some Euclidean space E .



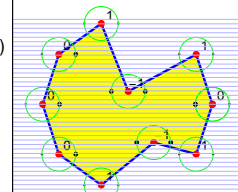
Embed the triangulation into space E .



Most height functions are Morse on M and colorings on G .

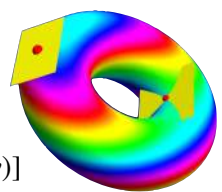
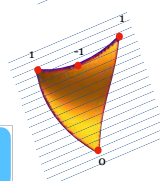
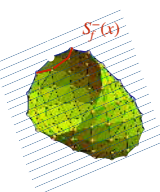
Nash-embed

Use rotational symmetric measure on sphere $S = \{v \mid |v|=1\}$



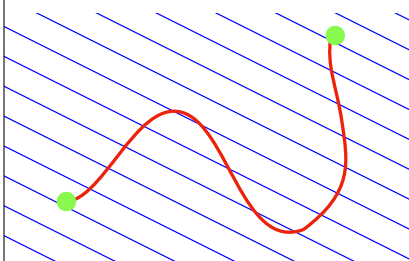
Nash embed M into Euclidean space E . Use linear functions

$$g_v(x) = x \cdot v$$



$K(v) = E[i_g(v)]$
 is GBChern integrand

Integral Geometry



Prototype result of Crofton:
 The length of a curve is the proportional to the expected number $E[i_\alpha]$ of intersections

$$i_\alpha = |\{g_\alpha(x) \in \mathbb{Z}\}|$$

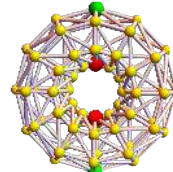
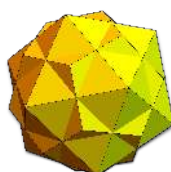
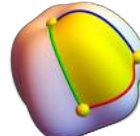
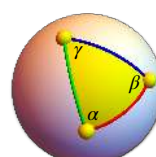
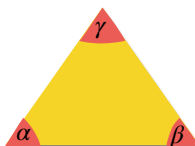
with an equidistant parallel grid. See Buffon needle.

Morgan Crofton
 (1826-1915)



Curvature Flavors

Any curvature in differential geometry can be written integral geometrically!



1827 Gauss



1848 Bonnet



1888 von Dyck



1940 Allendoerfer/Fenchel



1927 Hopf



1943 Allendoerfer-Weil



1944 Chern

